

spikeEncode — decisions needed from Oliver

9 September 2026 · companion to the encoder survey, version 3 · commit de6dee5

Where the work is. All six encoders implemented, known-answer suite green, and the harness running all three tasks end to end with the non-spiking reference alongside — weeks 1–2 and 5–7 of the plan, at day 21.

What just changed. The LDC account is accepted, so O2 — the largest blocker for three weeks — is clearing and stage one starts on TIMIT as soon as the data is in hand (D81). Until then every number is on a synthetic stand-in, whose known ground truth is what lets the pipeline controls actually fail — and they have, repeatedly. What it cannot do is evaluate control C1, the one saying the pipeline is calibrated and not merely self-consistent. **Nothing below blocks the next fortnight:** T1 and T3 run the day the data lands, and T2 only waits on a pitch tracker, since TIMIT carries no f₀ reference and §4.2 wants two of them.

Survey v3 replaces every task figure in the copy you have. v2's free parameters — chiefly the label alignment — were chosen by maximising the score on the *test* split, which the protocol forbids. All five task results are re-run with them chosen inside the training split. **T1 is unaffected: the bias was exactly zero.** T2, T3 without context and P1 are not, and P1's index changes sign. Nothing about the encoders changed — only which number picked the settings. §13.4 measures it.

	Decision	What it is holding up
Q43	The corpus question you raised — TIMIT despite the accent mismatch?	Answered as D81 in the reply; worth your confirmation. Nothing here is pretrained, so only a <i>ranking</i> transfers, and testing whether it survives an accent change is what stage two is for. Neither open corpus has hand-placed phone labels — T1's labels, T3's truth, and the aligner's calibration.
Q07	Release format — are ON and OFF separate channel indices, or a polarity bit?	The event file format, and controls C6 and C7 with it. Best settled before the featurisation is fixed, since doubling the channel count changes what the decoder sees. Interacts with the SHD convention, which is unipolar.
O3	Spiketrum — who to approach, and on what terms.	The primary source has been read and it changes the question (D80). The work is led from Zhejiang by Tang and colleagues, who are corresponding; Wijekoon and Alsakkal are two of eight. Survey v2 said it was developed at Manchester by those two — that was wrong , and it matters because the approach may need to go to Tang. The algorithm is fully published and needs no code from them, so what is at stake is attribution, not feasibility: may we implement it and report it as <i>our</i> implementation of their algorithm?
Q31	The mel reference does not bound T3, and v3 widens the gap.	E1 scores F 0.7557 against the reference's 0.5930, and the gap <i>grew</i> under the correction above. Either the reference is mis-specified for a timing task, or §5.9 is wrong to call it an upper bound. Decides how every T3 figure is reported.

Not blocking today: O1 corpus specifics, needed before stage two; O4 whether an IOP read-and-publish agreement covers the article charge; O5 alignment strategy, which depends on O1. O2 is no longer listed: the LDC account is accepted. Q28, whether to make the stand-in more speech-like, is largely settled by that — P1 cannot be answered on it either way. 20 questions are open against the design session, nine of them waiting on a single patch; only the rows above need you.

One result worth a minute of the meeting. The study's central engineering question has its first number and it does not favour the answer the field assumes. The mel reference costs 128,000 bit/s; the spiking encoder reaches 98.5 per cent of its accuracy only at **398,929 bit/s, three times the representation it is approximating**, crossing near a fifth of that rate at ~91 per cent of the bound. Declared bit widths, synthetic corpus — not yet a result, but the shape of the argument the paper will have to make.